

```
1  use OnlineBookstore;
2
3  show tables;
4
5  Select * from books;
6  Select * from customers;
7  Select * from orders;
8
9  Show columns from books;
10 Show columns from customers;
11 Show columns from orders;
12
13 -- Adding Primary Key
14 Alter table books
15 add primary key (Book_ID);
16
17 Alter table customers
18 add primary key (Customer_ID);
19
20 Alter table orders
21 add primary key (Order_ID);
22
23 -- 1) Retrieve all books in the "Fiction" genre:
24 Select *
25 from books
26 where Genre = 'Fiction'
27 order by Title;
28
29 -- 2) Find books published after the year 1950:
30 Select *
31 from books
32 where Published_Year > 1950
33 Order by Published_Year;
34
35 -- 3) List all customers from the Canada:
36 Select *
37 from customers
38 where Country = 'Canada'
39 order by Name ASC;
40
41 -- 4) Show orders placed in November 2023:
42 Select *
43 from orders
44 where Order_Date between '2023-11-01' and '2023-11-30'
45 Order by Order_Date;
46
47 -- 5) Retrieve the total stock of books available:
48 Select Sum(Stock) as Total_Stocks
49 from books;
50
51 -- 6) Find the details of the most expensive book:
52 Select *
53 from books
54 order by Price DESC
55 Limit 1 ;
56
57 -- 7) Show all customers who ordered more than 1 quantity of a book:
58 Select *
59 from orders
60 where quantity > 1
61 order by quantity;
62
63 -- 8) Retrieve all orders where the total amount exceeds $20:
64 Select *
65 from orders
66 where Total_Amount > 20
67 Order by Total_Amount Desc;
68
69 -- 9) List all genres available in the Books table:
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70  Select Distinct Genre
71  from books
72  Order by Genre;
73
74  -- 10) Find the book with the lowest stock:
75  Select *
76  from books
77  order by stock
78  limit 1;
79
80  -- 11) Calculate the total revenue generated from all orders:
81  Select round(sum(total_amount),0)
82  from orders;
83
84  -- Advance Questions :
85
86  -- 1) Retrieve the total number of books sold for each genre:
87  Select B.Genre, sum(O.quantity) as Total_Books_Sold
88  from books B
89  Join orders O on B.book_id = O.book_id
90  group by B.Genre
91  Order by Total_Books_Sold DESC;
92
93
94  -- 2) Find the average price of books in the "Fantasy" genre:
95  Select Genre, round(Avg(Price),2) as Avg_Price
96  from books
97  where Genre = 'Fantasy'
98  order by Avg_Price;
99
100
101  -- 3) List customers who have placed at least 2 orders:
102  Select C.Name, C.customer_id, count(O.Order_id) as Order_Count
103  from Orders O
104  Join customers C on O.customer_id = C.customer_id
105  group by C.Name, C.customer_id
106  having count(O.Order_id)>=2
107  Order by Order_Count;
108
109  -- 4) Find the most frequently ordered book:
110
111  Select b.Title, b.Genre, o.Book_id, count(o.order_id) as Order_Count
112  from orders o
113  join books b on b.book_id = o.book_id
114  group by book_id
115  order by Order_Count Desc
116  Limit 10;
117
118
119  -- 5) Show the top 3 most expensive books of 'Fantasy' Genre :
120  Select Title, Genre, Price
121  from books
122  where Genre = 'Fantasy'
123  order by Price Desc
124  Limit 3;
125
126
127  -- 6) Retrieve the total quantity of books sold by each author:
128  Select b.Author, sum(o.quantity) as total_quantity_sold
129  from orders o
130  join books b on o.book_id = b.book_id
131  group by b.Author
132  order by total_quantity_sold DESC;
133
134  -- 7) List the cities where customers who spent over $30 are located:
135  Select c.City, o.Total_amount
136  from orders o
137  join customers c on o.customer_id = c.customer_id
138  where Total_Amount > 30

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139 order by Total_Amount DESC;
140
141 -- 8) Find the customer who spent the most on orders:
142 Select c.Name, c.customer_id, round(sum(o.Total_Amount),0) as Total_Amount_Spent
143 from orders o
144 join customers c on o.customer_id = c.customer_id
145 group by c.Name, c.customer_id
146 order by Total_Amount_Spent DESC
147 Limit 3;
148
149 -- 9) Calculate the stock remaining after fulfilling all orders:
150 Select b.book_id, b.title, b.stock, coalesce(sum(o.quantity),0) as Order_quantity,
151        b.stock - coalesce(sum(o.quantity),0) as Remaining_quantity
152 from books b
153 join orders o on b.book_id = o.book_id
154 group by b.book_id;
```